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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2024
END SEMESTER EXAMINATION - ODD 2024
HU104 - ENVIRONMENTAL SCIENCE

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A

(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) Minamata disease is related to - (1) CO2 BL2

- a) As
- b) Hg
- c) Cd
- d) Pb

1.ii) UNEP stands for - (1) CO3 BL1

- a) United Nations Environmental Policy
- b) United Nations Environmental Programme
- c) United Nations Environmental Protection
- d) All of these

1.iii) The term 'stone cancer' is associated with- (1) CO2 BL2

- a) acid rain
- b) photochemical smog
- c) Ozone depletion
- d) None of these

1.iv) The most potential renewable energy resource is in India (1) CO1 BL2

- a) Geothermal
- b) Wind
- c) Biomass
- d) Solar

1.v) Decibel is the unit of (1) CO4 BL1

- a) Intensity
- b) Wave length
- c) Frequescy
- d) None

- 1.vi) The full form of PAN is- (1) CO2 BL2
 a) Peroxy alkyl nitrate
 b) Peroxy acyl nitrate
 c) Peroxy acyl nitrite
 d) Peroxy alkyl nitrite
- 1.vii) For a wastewater sample, COD is generally _____ than BOD. Fill in the blank from the options given. (1) CO1 BL1
 a) greater
 b) lower
 c) equal
 d) can not be predicted
- 1.viii) Ozone is a pollutant when present in (1) CO2 BL2
 a) Troposphere
 b) Stratosphere
 c) Mesosphere
 d) Thermosphere
- 1.ix) Population Growth depends on (1) CO1 BL2
 a) Death rate
 b) Birth rate
 c) Migration rate
 d) All of these
- 1.x) Lakes rich in nutrients are known as (1) CO3 BL3
 a) Mesotrophic lake
 b) Oligotrophic lake
 c) Eutrophic lake
 d) Atrophic lake
- 1.xi) Synthetic azo dyes are examples of- (1) CO2 BL2
 a) Non-biodegradable
 b) Biodegradable
 c) Partially biodegradable
 d) None of these
- 1.xii) The National Institute of Disaster Management (NIDM) was established on _____ (1) CO4 BL2
 a) September 23, 2002
 b) August 14, 2001
 c) October 16, 2003
 d) August 14, 2004

Group B

(Short Answer Type Questions)

(Answer any three of the following) 3x5=15

2. Write a brief note about the following water quality parameters: i) TDS ii) Alkalinity [2.5+2.5] (5) CO2 BL3

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|--|-----|-----|-----|
| 3a. Write the equation for exponential population growth and explain the notations. | (3) | CO1 | BL3 |
| 3b. How to calculate the "Natural Increase Rate"? | (2) | CO1 | BL2 |
| 4a. What do you mean by municipal waste composting? | (3) | CO2 | BL2 |
| 4b. Why is biomedical waste considered as hazardous waste? | (2) | CO2 | BL3 |
| 5. A wastewater sample has an ultimate BOD of 280 mg/l and a 5-day BOD of 240mg/lt. Calculate the 20-day BOD of this sample. | (5) | CO4 | BL4 |
| 6a. What are the different sources of noise pollution? | (3) | CO2 | BL2 |
| 6b. Explain the purpose of $L_{10}(18)$ index w.r.t noise pollution. | (2) | CO2 | BL2 |

Group C

(Long Answer Type Questions)

(Answer any three of the following) 3x15=45

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| 7. Answer the following questions:
(a) What is global warming? [3]
(b) How does greenhouse gases cause global warming? [6]
(c) What are the adverse effects of global warming? [6] | (15) | CO2 | BL3 |
| 8. Answer the following:
(a) What do you mean by Ecosystem? [3]
(b) What are the components of the Ecosystem? [6]
(c) Write down the oxygen cycle with a diagram. [6] | (15) | CO1 | BL2 |
| 9. (a) Define disaster management. What are the different types of disasters? [2+3]
(b) Explain the disaster management cycle in detail. [5]

(c) How can vulnerability to disasters be assessed and characterized? [5] | (15) | CO4 | BL3 |
| 10. Write short notes on:
a) Cyclone separator [5]

b) Environmental Auditing [5]

C) E-waste management [5] | (15) | CO3 | BL3 |
| 11a. What is acid rain? [1]
Write down the causes, reactions and effects of acid rain. [4] | (5) | CO2 | BL2 |
| 11b. What do you mean by IR active molecules? Define atmospheric window | (5) | CO2 | BL3 |
| 11c. Discuss about the advantages of land filling | (2) | CO4 | BL1 |
| 11d. Write down the each components of Forest eco-systems | (3) | CO1 | BL1 |